

$$\frac{\partial \left(\frac{A_1 \left(1 - \frac{1}{e^{-(\nu-\omega)} + 1} \right) + \frac{A_2}{e^{-(\nu-\omega)} + 1}}{d} - \nu + y_1 \right)}{\partial \nu} = \frac{\frac{A_2 e^{\omega-\nu}}{(e^{\omega-\nu} + 1)^2} - \frac{A_1 e^{\omega-\nu}}{(e^{\omega-\nu} + 1)^2}}{d} - 1$$

The first order conditions lead to $e^x = \frac{(A_2 - A_1)e^\omega}{2d} - e^\omega - e^\omega \sqrt{\left(\frac{A_2 - A_1}{2d} - 1\right)^2 - 1}$
or $e^{x-\omega} = \frac{(A_2 - A_1)}{2d} - 1 - \sqrt{\left(\frac{A_2 - A_1}{2d} - 1\right)^2 - 1}$
For the richer consumer,

$$\frac{\partial \left(\frac{A_1 \left(1 - \frac{1}{e^{-(\nu-\lambda)} + 1} \right) + \frac{A_2}{e^{-(\nu-\lambda)} + 1}}{d} - \nu + y_2 \right)}{\partial \nu} = \frac{(A_2 - A_1)e^{\lambda-\nu}}{d \times (e^{\lambda-\nu} + 1)^2} - 1$$

The first conditions for the richer consumer are similarly based on $\underline{\lambda}$.